Ecommerce – Order & Returns Analysis

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Tool Used: Excel (Data Cleaning & Validation), Power BI (Analysis & Dashboard)

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1 Executive Summary

Urban Threads experienced strong growth during Q4 2024, driven primarily by festive demand during the Diwali period. However, this growth was accompanied by a significant increase in return rates, impacting operational efficiency and profitability.
This analysis focuses on identifying the key drivers behind the high return rate and provides actionable recommendations to reduce returns without negatively impacting sales.

Data Overview & Preparation

Dataset Description

The dataset contains **346 orders** placed between **October–December 2024**, including order details, delivery status, returns, payment methods, and product categories.

Data Cleaning & Validation (Summary)

- Missing values were reviewed and handled based on business logic (e.g., return fields blank when no return was initiated).
- A spelling inconsistency in delivery_status was corrected.
- One invalid quantity value (-1) was identified and corrected.
- All validation rules provided in the assessment were checked and documented in a **Cleaning Log** and **Validation Summary**.

No critical data quality issues remain after cleaning.

Key Performance Metrics (Q4 2024)

Metric	Value
Total Orders	346
Total Revenue	₹1.02M
Average Order Value (AOV)	₹2,959
Delivered Orders	328
Overall Return Rate	~55%

Observation:
While revenue and order volume indicate strong demand, the return rate is significantly high, suggesting underlying operational or customer-experience issues.

Trend Analysis Weekly Order Volume Trend

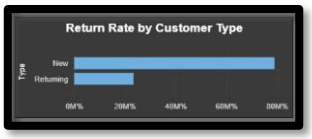


- Order volume increased steadily through the quarter.
- A noticeable spike occurs during festive weeks (Diwali period).
- This indicates strong promotional demand but also potential quality or expectation mismatches.

Insight:
Peak demand periods are likely contributing to higher downstream returns.

Segmentation Analysis

a) Return Rate by Customer Type

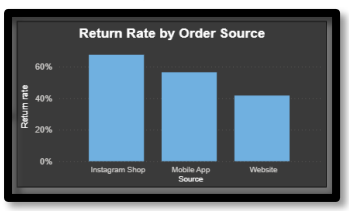


- **New customers** have a significantly higher return rate than returning customers.

Interpretation:
First-time buyers may have mismatched expectations related to sizing, quality, or product representation.

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b) Return Rate by Order Source

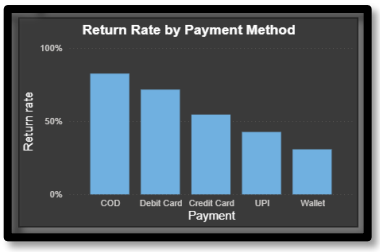


- **Instagram Shop** shows the highest return rate.
- Website and Mobile App perform comparatively better.

Interpretation:
Social commerce channels may attract impulse buyers with lower purchase intent or misaligned expectations.

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c) Return Rate by Payment Method

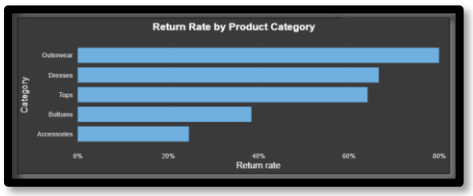


- **Cash on Delivery (COD)** orders have the highest return rate.
- Prepaid methods (UPI, Wallets) show lower return rates.

Interpretation:
COD orders carry higher risk due to lower customer commitment.

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d) Return Rate by Product Category



- Certain categories (e.g., Outerwear, Dresses, Tops) contribute disproportionately to returns.

Interpretation:
Fit, sizing, or quality perception may vary significantly across categories.

🔄 Business Interpretation

1. **Why is the return rate high?**
   - High share of new customers
   - Heavy reliance on COD orders
   - High-return channels like Instagram Shop
   - Category-specific sizing and quality issues
2. **What does this indicate?**
   - Growth is coming at the cost of operational inefficiency.
   - Returns are not evenly distributed but concentrated in specific segments.

Recommendations (CEO-Focused)

Goal: Reduce return rate from ~55% to below 15% within 2 quarters

Recommendation 1: Improve New Customer Experience

- Enhance size guides and product descriptions
- Add real customer images and reviews
- Introduce post-order education emails for first-time buyers

Expected Impact:
10–15% reduction in returns from new customers

Recommendation 2: Optimize COD Usage

- Restrict COD for high-risk segments (new users, Instagram Shop)
- Offer small incentives for prepaid orders

Expected Impact:
Lower RTO and return costs, improved delivery success

Recommendation 3: Fix Category-Specific Issues

- Audit high-return categories
- Improve sizing consistency and quality checks
- Pilot free alterations for size-related returns

Expected Impact:
Reduced size-related returns without impacting sales

📘 What I Would Test Next

A/B Test Proposal:

- Compare return rates for new customers with:
  - Enhanced size guidance vs. standard product pages

Additional Data Needed:

- Customer feedback / reviews
- SKU-level size complaints
- Repeat purchase behavior post-return

Assumptions & Limitations

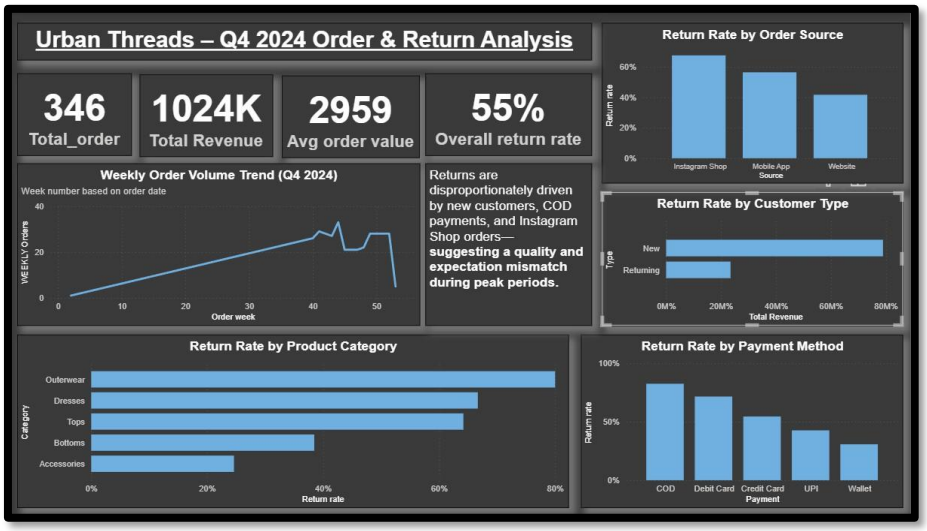
- Dataset represents only Q4 2024; long-term trends are not visible.
- No SKU-level defect or manufacturing data available.
- Some returns may be in-progress at the time of data capture.

Conclusion

Urban Threads has achieved strong growth, but return rates pose a serious profitability challenge. By focusing on **new customer experience**, **COD optimization**, and **category-level improvements**, the company can significantly reduce returns while maintaining revenue momentum.

Attachments

- # Dashboard



- # Cleaning Log

Column1	Column2	Column3	Column4
Rule ID	Validation Check	Status	Notes
V1	order_id follows UT-XXXXXX format	Pass	No duplicate or invalid order IDs found
V2	customer_type contains only New / Returning	Pass	No other values observed
V3	quantity is a positive integer	Pass	One negative value corrected during cleaning
V4	discount_percent between 0 and 100	Pass	All values within valid range
V5	order_date within Q4 2024 (Oct–Dec)	Pass	No dates outside required range
V6	payment_method from allowed list	Pass	No unexpected payment methods
V7	Delivered orders have delivery_date	Pass	Delivery logic validated
V8	return_initiated = No → return_reason blank	Pass	Matches business rules
V9	warehouse_origin valid (Mumbai/Delhi/Bangalore)	Pass	No invalid warehouse names
V10	refund_status logic consistency	Pass	No inconsistencies found

- # Validation Summary

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